

Subclinical Hypothyroidism in Polycystic Ovary Syndrome: Prevalence and Impact on Metabolic and Cardiovascular risk

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Taieb Ach^{1,2,3}, Rim Dhaffar¹, Fatma Ben Abdesslem^{1,2},
Wiem Saafi^{1,2}, Imen Halloul^{1,2}, Hamza ElFekih^{1,2}, Ghada Saad^{1,2}
and Yosra Hasni^{1,2}

Abstract

Background: Polycystic ovary syndrome (PCOS) is a complex condition linked to long-term health risks such as cardiovascular disease, type 2 diabetes, and metabolic syndrome. Subclinical hypothyroidism (SCH) shares overlapping symptoms with PCOS, but their relationship remains debated. SCH prevalence in PCOS patients and its impact on cardiovascular and metabolic health is debated and thus warrants further research. This research examined the association between SCH and PCOS in a Tunisian clinical-based population.

Methods: We conducted a prospective cross-sectional study of 161 PCOS patients and 75 age-matched controls recruited from 2023 to 2024. All participants underwent thyroid function testing, metabolic profiling, and hormonal assays. Statistical analyses included Mann-Whitney U, Kruskal-Wallis, and chi-square tests, with age/BMI-adjusted linear regression models.

Results: Among 236 participants (161 PCOS, 75 controls), SCH prevalence was higher in PCOS patients (14.9%) compared to controls (5.3%, $p = .048$). Adjusted for age and BMI, median TSH levels were higher in the PCOS group (2.2 mIU/L vs 1.75 mIU/L, $p = .006$), while FT4 levels were lower (14.26 pmol/L vs 15.26 pmol/L, $p = .007$). Positive TPOAb prevalence was higher in PCOS (12.4% vs 2.7%, $p = .049$). TSH levels varied across PCOS phenotypes ($p = .003$), with Phenotypes A and B showing higher levels than Phenotype C ($A > C$, $p = .019$; $B > C$, $p < .001$). SCH was highest in Phenotype A. SCH in PCOS was associated with impaired glucose tolerance ($p = .011$), higher blood fasting glucose ($p = .033$), higher total cholesterol ($p = .022$), and hypertriglyceridemia ($p = .018$).

Conclusion: SCH is more prevalent in PCOS and may worsen insulin resistance and dyslipidemia. Addressing thyroid dysfunction in PCOS patients may be beneficial for more effective management strategies, ultimately improving reproductive, metabolic, and cardiovascular outcomes for affected women.

Plain language summary

How Common is Mild Thyroid Problems in PCOS, and How Do They Affect Health Risks?

Polycystic Ovary Syndrome (PCOS) is a common condition that affects how a woman's ovaries work. It can cause symptoms like irregular periods, weight gain, and difficulty getting pregnant. Some women with PCOS also have a mild thyroid problem called subclinical hypothyroidism, where the thyroid gland doesn't work as well as it should, but not enough to cause obvious symptoms. This study looked at how common this mild thyroid problem is in women with PCOS and whether it increases their risk of developing other health issues, like diabetes, high cholesterol, or heart disease. The findings suggest that subclinical hypothyroidism is more common in women with PCOS than in those without it. Additionally, having both PCOS and a mild thyroid problem may increase the risk of metabolic and cardiovascular problems, such as higher blood sugar levels, weight gain, and a greater chance of heart disease. Understanding this link

¹Faculty of Medicine of Sousse, University of Sousse, Tunisia

²Department of Endocrinology, University Hospital of Farhat Hached Sousse, Tunisia

³Laboratory of Exercise Physiology and Pathophysiology, University Hospital of Farhat Hached Sousse, Tunisia

Corresponding Author:

Taieb Ach, Department of Endocrinology, University Hospital of Farhat Hached Sousse, 17 Rue SH, Sousse 4000, Tunisia.

Email: ach.taieb@gmail.com



is important because it could help doctors better manage PCOS and reduce the risk of long-term health problems. Women with PCOS might benefit from regular thyroid checks and early treatment if needed. This research highlights the need for more awareness and tailored care for women with PCOS to improve their overall health.

Keywords

polycystic ovary syndrome, subclinical hypothyroidism, thyroid-metabolic-risk

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Introduction

Polycystic ovary syndrome (PCOS) is a complex, heterogeneous disorder affecting 8% to 13% of women of reproductive age, and up to 70% of cases go undiagnosed.¹ PCOS is associated with long-term health complications, including impaired glucose tolerance, type 2 diabetes, metabolic syndrome, and cardiovascular conditions.² The syndrome's pronounced clinical variability and unclear pathophysiology pose considerable challenges for clinicians in achieving effective management,³ emphasizing the need for precise and personalized treatments. While polycystic ovarian morphology is 1 component of Rotterdam criteria, PCOS diagnosis requires careful exclusion of other endocrinopathies and documentation of either hyperandrogenism or ovulatory dysfunction.¹

Subclinical hypothyroidism (SCH) and overt hypothyroidism, common endocrine disorders,^{4,5} are primary caused by iodine deficiency and autoimmune conditions.^{6,7} A link between PCOS and autoimmune disorders, such as Hashimoto's thyroiditis, has been observed, with anti-thyroid peroxidase antibodies (anti-TPO) frequently detected.⁸ Despite their etiopathogenesis being completely different, thyroid disorders often mimic PCOS,⁵ sharing features like menstrual irregularities, infertility, and metabolic disturbances.⁹ These overlapping characteristics highlights the need for thorough evaluation to ensure accurate diagnosis and management. SCH, defined as elevated thyroid-stimulating hormone (TSH) with normal free thyroxine (FT4) levels,¹⁰ affects about 10% of the general population but is less common in reproductive-aged women, with a prevalence of 4% to 6%.^{11,12} However, SCH prevalence is significantly higher in PCOS patients, ranging from 10% to 25%.⁹ While evidence supports a connection between SCH and PCOS,^{13,14} this relationship remains debated due to conflicting findings in the literature.¹⁵⁻¹⁸ Furthermore, evidence suggests a connection between SCH and elevated cardiac risks, metabolic syndrome, dyslipidemia, and insulin resistance.¹⁹⁻²¹ However, it remains uncertain whether SCH directly increases the likelihood of cardiovascular diseases.²² There have been several studies investigating the effects of SCH on PCOS women's metabolic and hormonal outcomes.²³⁻²⁵ There is, however, no firm consensus in this field based on current data.

Given the potential influence on fertility, metabolism, and overall hormonal balance, studying the relationship between SCH and PCOS is particularly relevant as this could improve the management of patients with coexisting conditions. In the present work, we aim to explore the association between SCH and PCOS in a clinical-based population and to investigate the impact of SCH on anthropometric, lipid, glucose, and hormonal profiles of patients in PCOS patients.

Patients and Methods

Study Population and Design

This prospective cross-sectional observational study was conducted in University Hospital of Farhat Hached Sousse Tunisia, setting between 2023 and 2024, including patients with PCOS and a control group.

The sample size was calculated based on the formula that is specific to our type of study, that is:

$$n = \frac{z^2 \cdot p \cdot (1-p)}{m^2},$$

with:

n = sample size,

z = confidence level: for a 95% confidence level, $z = 1.96$,

p = prevalence of PCOS among women of reproductive age (15-45 years) in North Africa in 2019, estimated at 4.73%,

m = the tolerated margin of error or precision; we aim to ascertain the true proportion within a 5% margin.

The calculation of the formula yields a sample size $n = 70$ patients. This threshold will be statistically representative of the PCOS population. The final number was increased, given the increase in the prevalence of PCOS in Tunisia.

Participants were recruited during medical consultations in the outpatient endocrinology department, and their clinical, hormonal, and metabolic data were collected at the time of the study. PCOS diagnosis was based on Rotterdam criteria, after excluding other causes of hyperandrogenism or menstrual irregularities. Additionally, patients who had received hormonal treatment that could interfere with hormone assays or ultrasound evaluations within the past 3 months were not included. Other exclusion criteria included hormonal treatment within the prior 3 months, chronic kidney, liver, or heart failure, active or past neoplastic disease, gonadotoxic therapy, pregnancy/lactation, postmenopausal status, primary amenorrhea, and transient causes of elevated thyrotropin, such as thyroiditis recovery or medications (eg, amiodarone, lithium). PCOS patients were further classified into 4 phenotypes based on diagnostic features.²⁶ A control group was included to assess SCH prevalence. It consisted of 75 age-matched women who presented to outpatient clinics for reasons unrelated to PCOS or thyroid disorders and had regular menstrual cycles. Controls were recruited from women attending outpatient clinics for routine gynecological exams, contraceptive counseling, minor dermatological conditions. All had regular menstrual cycles, no hyperandrogenism or thyroid disorders, and were

Table 1. Baseline characteristics of PCOS and control groups. Data are presented as medians with interquartile ranges (IQR).

Clinical features	Overall (n = 236)	PCOS (n = 161)	Controls (n = 75)	p-Value
Age	22 [IQR = 20-26]	22 [IQR = 20-26]	22 [IQR = 21-26]	.386
BMI (kg/m ²)	26.88 [IQR = 23.34-32.61]	29 [IQR = 24.24-34]	24.53 [IQR = 22.96-28]	<.001
WHR	0.75 [IQR = 0.72-0.82]	0.76 [IQR = 0.72-0.82]	0.74 [IQR = 0.71-0.79]	.341

Abbreviations: BMI, body mass index; PCOS, polycystic ovary syndrome; WHR, waist to hip ratio.

medication-free. We excluded those with endocrine/metabolic complaints or chronic diseases.

Clinical and metabolic data were collected prospectively through structured physician-administered questionnaires covering medical history and symptoms, standardized anthropometric measurements performed by trained staff, and laboratory analyses of fasting blood samples drawn between 8 and 10 AM following overnight fasting. All data were recorded electronically using case report forms with automated range checks.

To evaluate SCH's impact on hormonal and metabolic profiles, PCOS patients were divided into SCH-PCOS and Euthyroid-PCOS groups based on their most recent TSH levels.

Biochemical Assay

Blood samples were collected during standardized endocrinological screenings and follow-up. Blood samples were collected uniformly between 8:00 and 10:00 AM after a 12-hour overnight fast for all participants, with serum processed within 2 hours using standardized platforms (ELISA for TPOAb). Hormonal assay aliquots were stored at -80°C until batch analysis to ensure consistency. All blood samples were collected during the early follicular phase (cycle days 2-5) for menstruating participants. For women with oligo/amenorrhea, sampling was performed after progesterone-induced withdrawal bleeding or following ≥ 2 months of spontaneous amenorrhea.

Women with thyroid dysfunction medication use were excluded from TSH and FT4 comparisons. Serum TSH levels were measured using electrochemiluminescence immunoassay (ECLIA) on the Cobas e801 analyzer (Roche Diagnostics).

SCH was defined as TSH > 4.5 mIU/L with normal FT4.²⁷ TPOAb levels were measured using ELISA (Euroimmun®), with a positivity threshold of 50 IU/mL. Fasting blood glucose (FBG) was assessed using the glucose oxidase method, with categories defined by ADA criteria.²⁸ HbA1c $\geq 6.5\%$ indicated diabetes, and 5.7% to 6.4% indicated prediabetes.²⁹ For high-density lipoprotein cholesterol (HDL-C), the ideal value is greater than 0.55 g/L in women. Low-density lipoprotein cholesterol (LDL-C) was evaluated using the Friedewald formula.³⁰ The optimal blood TG level in women should be < 1.20 g/L (1.3 mmol/L).

Statistical Analyses

Data were analyzed using parametric or nonparametric tests based on distribution. Continuous variables were expressed as medians and IQR, and categorical variables as numbers

and percentages. Mann-Whitney *U* and Kruskal-Wallis tests were used for continuous data, while chi-square or Fisher's exact tests were used for categorical data. Factors with the potential to affect PCOS (age and BMI) were adjusted and included using linear regression models. To examine whether there was a statistically significant difference between the medians of the 4 phenotypes, we performed a Kruskal-Wallis test. A Bonferroni correction was applied for multiple comparisons. Analyses were performed using SPSS version 25, with $P < .05$ considered significant.

Results

Patients Baseline Characteristics

A total of 236 participants were included, comprising 161 women with PCOS and 75 controls. The baseline characteristics, including age, body mass index (BMI), and waist-to-hip ratio (WHR), are summarized in Table 1. Age was comparable between the 2 groups, with a median of 22 years for both (PCOS: IQR 20-26; controls: IQR 21-26; $P = .386$). Similarly, no significant difference was observed in WHR between the PCOS group (median 0.76, IQR 0.72-0.82) and the control group (median 0.74, IQR 0.71-0.79; $P = .341$). However, BMI was significantly higher in the PCOS group (median 29, IQR = 24.24-34) compared to the control group (median 24.53, IQR = 22.96-28; $P < .001$).

Thyroid Function in PCOS Group and Controls

Among PCOS patients, 85.1% were euthyroid, whereas 14.9% had SCH. In comparison, 94.6% of controls were euthyroid, and only 5.3% had SCH (Figure 1). This difference was statistically significant after adjusting for age and BMI ($P = .048$).

The median TSH levels were significantly higher in the PCOS group (2.2 mIU/L; IQR = 1.56-3.51) compared to the control group (1.75 mIU/L; IQR = 1.22-2.33; $P = .001$, adjusted $P = .006$). Similarly, the median FT4 levels were significantly lower in PCOS patients (14.26 pmol/L; IQR = 12.26-15.26) compared to controls (15.26 pmol/L; IQR = 12.54-16.3; $P = .01$, adjusted $P = .007$).

The prevalence of positive TPOAb was higher in the PCOS group (12.4%) compared to controls (2.7%). This difference was statistically significant ($P = .016$, adjusted $P = .049$). Among those with positive TPOAb, 17 PCOS patients and 1 control were diagnosed with SCH, while 3 PCOS patients and 1 control were euthyroid. Table 2 provides a detailed summary of these differences in thyroid function and hormonal levels between women with PCOS and controls.

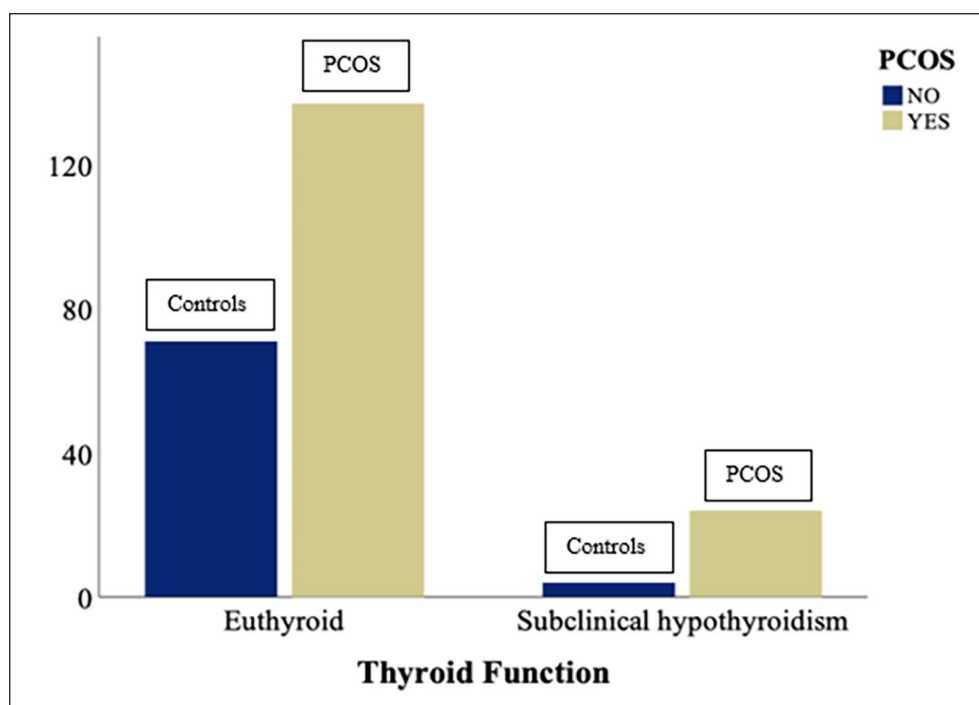


Figure 1. Distribution of thyroid function in PCOS and control groups. The bars represent the number of euthyroid and SCH cases in both PCOS and control groups.
Abbreviations: PCOS, polycystic ovary syndrome.

Table 2. Thyroid function and hormonal levels in PCOS and control groups. Data are presented as medians with interquartile ranges (IQR) or as percentages. Adjusted *P*-values account for BMI and age.

Thyroid aspects	PCOS (n = 161)	Controls (n = 75)	<i>P</i> -Value	Adjusted <i>P</i> -value
Thyroid function				
Euthyroid %	85.1	94.6	.034	.048
SCH %	14.9	5.3	.024	.04
TSH (mUI/L)	2.2 [IQR = 1.56-3.51]	1.75 [IQR = 1.22-2.33]	.001	.006
FT4 (pg/mL)	14.26 [IQR = 12.26-15.26]	15.26 [IQR = 12.54-16.3]	.01	.007
Positive TPOAb %	12.4	2.7	.016	.049
Euthyroid n (%)	3 (1.9)	1 (1.3)		
SCH n (%)	17 (10.6)	1 (1.3)		

Abbreviations: FT4, free thyroxine; PCOS, polycystic ovary syndrome; SCH, sub clinical hypothyroidism; TPOAb, anti-thyroid peroxidase antibodies; TSH, thyroid-stimulating hormone.

Table 3. TSH levels in PCOS phenotypes. Medians with interquartile ranges (IQR) are shown for Phenotypes A, B, and C, while exact values are provided for Phenotype D (n = 3). Post-hoc Bonferroni-adjusted comparisons were conducted for Phenotypes A, B, and C.

PCOS phenotypes	TSH (mUI/L) (Median)	<i>P</i> -Value	Post-hoc significance
Phenotype A	2.22 [IQR = 1.51-3.81]	0.003 (Kruskal-Wallis)	A > C (<i>P</i> = .015) (Bonferroni adjusted)
Phenotype B	3 [IQR = 2.18-4]		B > C <i>P</i> < .001 (Bonferroni adjusted)
Phenotype C	1.31 [IQR = 1.79-2.27]		A versus B (<i>P</i> = .084)
Phenotype D	2.17, 3.18, 4.41		

Abbreviations: TSH, thyroid-stimulating hormone.

Thyroid Function and PCOS Phenotypes

Among 161 PCOS patients, the distribution across phenotypes was as follows: Phenotype A (60.2%), Phenotype B (16.8%), Phenotype C (21.1%), and Phenotype D (1.9%).

The comparison of BMI across the 3 PCOS phenotypes revealed no statistically significant differences

(Kruskal-Wallis $H=0.301$, $P=.960$). BMI levels did not differ significantly across phenotypes (Kruskal-Wallis $H=0.301$, $P=.960$), with medians ranging from 23.34 to 29.19 kg/m².

TSH levels differed significantly among phenotypes ($P=.003$), as summarized in Table 3. Excluding phenotype D, post-hoc pairwise comparisons revealed that Phenotypes

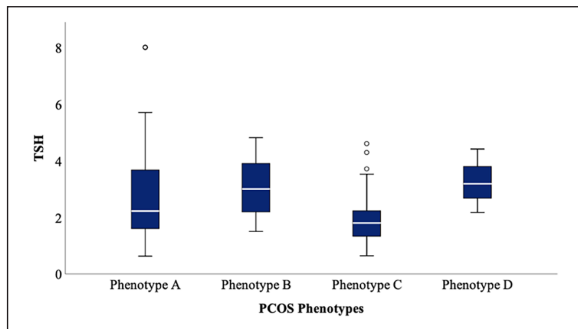


Figure 2. TSH levels in PCOS phenotypes. Box plots represent median, interquartile range, and outliers 9 (dots). Abbreviations: TSH (measured in mIU/L), thyroid-stimulating hormone; PCOS, polycystic ovary syndrome; Phenotype A, hyperandrogenism + ovulatory dysfunction + polycystic ovaries; Phenotype B, hyperandrogenism + ovulatory dysfunction; Phenotype C, hyperandrogenism + polycystic ovaries; Phenotype D, ovulatory dysfunction + polycystic ovaries.

A and B had significantly higher TSH levels than Phenotype C ($A > C$, $p = .019$; $B > C$, $p < .001$). Phenotype B exhibited the highest median TSH (3.0 mIU/L, IQR=2.18-4), followed by Phenotype A (2.22 mIU/L, IQR=1.51-3.81). No significant difference was observed between Phenotypes A and B ($p = .084$). Figure 2 illustrates these findings.

In contrast, FT4 levels did not significantly differ among the phenotypes (Kruskal-Wallis $H=4.621$, $P=.141$). Median FT4 levels were as follows: 14.25 pmol/L (IQR=12.45-15.31) for Phenotype A, 14.26 pmol/L (IQR=11-15) for Phenotype B, and 14.53 pmol/L (IQR=13.12-16.22) for Phenotype C. FT4 levels for Phenotype D were 10, 13, and 15 pmol/L.

Phenotypic Characteristics in SCH and Euthyroid PCOS Groups

We compared hyperandrogenism (HA), ovulatory dysfunction (OD), polycystic ovarian morphology (PCOM), and PCOS phenotypes between the SCH ($n=24$) and euthyroid ($n=137$) PCOS groups (Table 4). OD was significantly more prevalent in the SCH group (87.5%) compared to the euthyroid group (65%; $p=.029$). No significant differences were observed in biochemical HA ($p=.614$), clinical HA ($p=.150$) or PCOM prevalence ($p=.988$). The distribution of PCOS phenotypes differed significantly between the SCH and euthyroid groups ($P=.047$). When classified by phenotypes: Phenotype A was more common in the SCH group (79.2%) compared to the euthyroid group (56.9%; $p=.038$). Other phenotypes (B, C, and D) showed no significant differences. Due to the small sample size for Phenotype D ($n=3$), findings related to this phenotype should be interpreted with caution. The distribution of SCH subjects across phenotypes is illustrated in Figure 3.

SCH and PCOS: Hormonal Parameters

The hormonal characteristics of women with PCOS, stratified by thyroid function, are summarized in Table 5.

Table 4. Phenotypic characteristics in SCH and euthyroid PCOS groups. Data are presented as percentages.

Features		SCH (n=24)	Euthyroid (n=137)	P-value
Biochemical HA	No	16.7	21.2	0.614
	Yes	83.3	78.8	
Clinical HA	No	0	8	0.150
	Yes	100	92	
OD	No	12.5	35	0.029
	Yes	87.5	65	
PCOM	No	16.7	16.8	0.988
	Yes	83.3	83.2	
Phenotypes				0.047
A: HA-OD-PCOM		79.2	56.9	0.038
B: HA-OD		16.7	16.8	0.957
C: HA-PCOM		4.2	24.1	0.082
D: OD-PCOM		0	2.2	0.522

Abbreviations: HA, hyperandrogenism; OD, ovulatory dysfunction; PCOM, polycystic ovarian morphology; SCH, subclinical hypothyroidism.

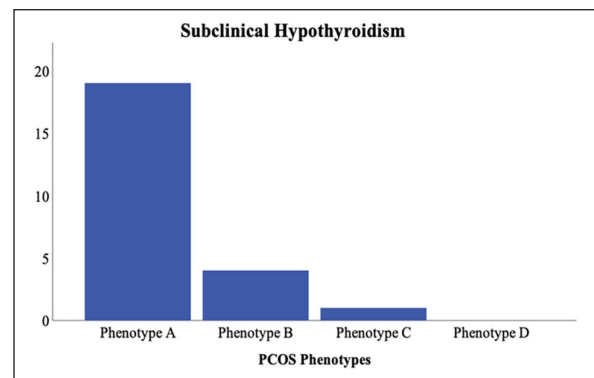


Figure 3. Frequency of subclinical hypothyroidism across PCOS phenotypes. The bars represent the number of SCH cases within each phenotype group.

Abbreviations: PCOS, polycystic ovary syndrome; Phenotype A, hyperandrogenism + ovulatory dysfunction + polycystic ovaries; Phenotype B, hyperandrogenism + ovulatory dysfunction; Phenotype C, hyperandrogenism + polycystic ovaries; Phenotype D, ovulatory dysfunction + polycystic ovaries.

Luteinizing hormone (LH) levels were significantly higher in the SCH group (median: 9.02, IQR: 9.02-17) compared to the euthyroid group (median: 9.04, IQR: 8.53-9.04; $p=.043$). Similarly, estradiol levels were significantly elevated in the SCH group (median: 64 pg/mL, IQR: 60-114) compared to the euthyroid group (median: 56 pg/mL, IQR: 48-69; $p=.004$). Other hormonal parameters, including Follicle-Stimulating Hormone (FSH) levels ($p=.745$), LH/FSH ratio ≥ 1 ($p=.231$) and free testosterone levels ($p=.317$), did not show significant differences.

SCH and PCOS: Clinical and Metabolic Parameters

No significant differences were observed in anthropometric characteristics, including age ($p=.294$), BMI ($p=.814$), and WHR ($p=.274$). Blood pressure (systolic: $P=.785$;

Table 5. Clinical, metabolic and hormonal characteristics in euthyroid and SCH subjects with PCOS. Data are presented as medians with interquartile ranges (IQR) or as percentages.

Biological items	SCH (n = 24)	Euthyroid (n = 137)	P-Value
LH (mIU/mL)	10 [IQR = 9.02-17]	9.04 [IQR = 5.83-9.04]	0.043
FSH (mIU/mL)	6 [IQR = 5.73-6.72]	6 [IQR = 5-7.49]	0.745
LH/FSH ratio ≥ 1 (%)	95.8	73.7	0.231
Free Testosterone (ng/mL)	0.72 [IQR = 0.60-0.85]	0.69 [IQR = 0.53-0.82]	0.317
Estradiol (pg/mL)	64 [IQR = 60-114]	56 [IQR = 48-69]	0.004
BMI (kg/m ²)	28.79 [IQR = 23.18-35.5]	29.06 [IQR = 24.17-34.21]	0.814
WHR	0.78 [IQR = 0.70-0.87]	0.75 [IQR = 0.72-0.82]	0.274
BP-SYS	11.5 [IQR = 11-13]	12 [IQR = 11-13]	0.785
BP-DYS	7 [IQR = 6-7.75]	7 [IQR = 6-7]	0.581
FBG (g/L)	1.02 [IQR = 0.86-1.1]	0.93 [IQR = 0.88-1]	0.033
HbA1c (%)	5.2 [IQR = 4.9-5.3]	5.2 [IQR = 5.1-5.55]	0.435
IGT (%)	54.2	26.3	0.011
TG (g/L)	1.05 [IQR = 0.82-1.68]	1.01 [IQR = 0.77-1.3]	0.274
Hypertriglyceridemia (%)	41.7	19.7	0.018
Cholesterol (g/L)	2.02 [IQR = 1.89-2.12]	1.88 [IQR = 1.7-2.02]	0.022
Hypercholesterolemia (%)	12.5	4.4	0.110
HDL-C (g/L)	0.52 [IQR = 0.46-0.60]	0.49 [IQR = 0.42-0.59]	0.324
Low-HDL-C (%)	41.7	46.7	0.647
LDL-C (g/L)	1.19 [IQR = 0.86-1.45]	1.06 [IQR = 0.91-1.23]	0.165
High LDL-C (%)	12.5	8.0	0.473

Abbreviations: BMI, body mass index; BP-DYS, diastolic blood pressure; BP-SYS, systolic blood pressure; FBG, fasting blood glucose; FSH, follicle stimulating hormone; HbA1c, glycated hemoglobin; HDL-C, high-density lipoprotein cholesterol; IGT, impaired glucose tolerance; IQR, interquartile range; LDL-C, low-density lipoprotein cholesterol; LH, luteinizing hormone; PCOS, polycystic ovary syndrome; SCH, subclinical hypothyroidism; TG, triglycerides; WHR, waist-to-hip ratio.

diastolic: $P = .581$) was also comparable between the 2 groups. Regarding the metabolic profile, HbA1c levels ($P = .435$), LDL cholesterol levels ($P = .165$), and HDL cholesterol levels ($P = .324$) showed no significant differences. However, fasting blood glucose (FBG) levels were significantly higher in the SCH group (1.02 g/L, IQR: 0.86-1.1) compared to the euthyroid group (0.93 g/L, IQR: 0.88-1; $P = .033$). The prevalence of impaired glucose tolerance (diabetes or prediabetes) was also significantly greater in the SCH group (54.2%) compared to the euthyroid group (26.3%; $P = .011$). Lipid profile analysis revealed significantly higher total cholesterol levels in the SCH group (2.02, IQR: 1.89-2.12) compared to the euthyroid group (0.93, IQR: 0.88-1; $P = .022$). Hypertriglyceridemia was more prevalent in the SCH group (41.7%) than in the euthyroid group (19.7%; $P = .018$). Table 5 provides a detailed comparison of these clinical and metabolic characteristics.

Discussion

Our findings indicated significant differences in the prevalence of serum TSH, serum FT4 levels, and TPOAb positivity between women with PCOS and controls. Specifically, SCH was more prevalent in PCOS patients compared to controls. These findings are consistent with a meta-analysis of 6 studies involving 692 women with PCOS and 540 controls, which demonstrated that women with PCOS have a more than threefold higher likelihood of developing SCH compared to those without PCOS when the analysis was

restricted to studies using a TSH cutoff of ≥ 4 mIU/L for SCH diagnosis.¹⁵ A further study by Glintborg et al demonstrated that women with PCOS had a significantly higher event rate of thyroid disease compared to controls, with an incidence rate of 6.0 per 1000 patient-years in PCOS versus 2.4 per 1000 patient-years in controls. The hazard ratio for thyroid disease development was 2.5 (95% CI 2.3-2.7).³¹ However, a recent population-based study by Rojhani et al¹⁷ involving 644 healthy controls and 207 PCOS-affected women revealed no correlation between PCOS status and SCH and no difference in 95 percentiles of TSH levels.³⁹

Differences in SCH prevalence across studies may stem from varying TSH cutoff levels.^{32,33} While a threshold of TSH > 4.5 mIU/L is commonly recommended, adverse pregnancy outcomes have been linked to levels > 2.5 mIU/L.³⁴ Consequently, a lower cutoff (TSH > 2.5 mIU/L) is often preferred for infertility or pregnancy planning, as adopted by Fatima et al.³⁵

The higher TSH levels were observed in a study by Sinha et al, where they found a significantly higher prevalence of autoimmune thyroiditis and elevated TSH levels in PCOS patients compared to controls. This suggests a subclinical hypothyroid state in PCOS. Most existing studies have explored the mean TSH levels in women with PCOS compared to controls, yielding inconclusive findings. While some studies reported elevated TSH levels in PCOS patients compared to controls,³⁶⁻³⁹ others found no significant differences between the 2 groups.⁴⁰⁻⁴³ These variations may

partially be attributed to differing approaches in accounting for confounding factors like BMI, as higher BMI has been directly associated with increased TSH levels.^{44,45}

The prevalence of TPOAb positivity in PCOS patients in this study further underscores the association between PCOS and thyroid dysfunction. TPOAbs support an autoimmune cause of SCH and are associated with a risk of progression to overt hypothyroidism approximately twice as high as when the test for thyroid peroxidase antibodies is negative.^{46,47} Unfavorably, some research has shown that women with PCOS are more likely to develop autoimmune thyroid disorders.⁴⁸ The Janssen et al⁴⁹ study indicates that the prevalence of autoimmune thyroiditis in these women is even 3 times higher.⁴⁹ According to other research, women with PCOS have higher serum TPOab levels than women without PCOS.^{48,50} A meta-analysis of 13 clinical-based studies conducted by Romitti et al,⁴⁴ encompassing a total of 1210 women with PCOS and controls, concluded that autoimmune thyroid disease occurs more frequently in women with PCOS compared to those without the condition.⁵¹ The connection between PCOS and thyroid dysfunction could be partially attributed to shared autoimmune mechanisms involved in both conditions. Autoimmune thyroiditis is the leading cause of SCH,⁵² while autoimmunity may also contribute to PCOS development, potentially through an imbalance between estrogen, and progesterone. Current evidence shows that progesterone offers a protective function as a natural immune suppressor, while estrogen stimulates autoimmune diseases.^{53,54} It has been proposed that women with PCOS may develop more autoimmune diseases as a result of lower progesterone levels.^{50,55}

When comparing thyroid function across PCOS phenotypes, significant differences in TSH levels were observed, with phenotypes A, and B exhibiting higher TSH levels compared to phenotype C. This suggests that thyroid dysfunction may manifest differently across PCOS phenotypes, potentially influenced by varying degrees of hyperandrogenism and ovulatory dysfunction. Similar patterns have been reported in other studies, where a notable association between PCOS phenotypes and TSH levels was identified. Higher TSH levels were linked to an increased prevalence of the complete PCOS phenotype A, characterized by hyperandrogenism, ovulatory dysfunction, and PCOM. Conversely, the prevalence of non-PCOM and non-hyperandrogenic phenotypes decreased as TSH levels rose. No significant impact was observed on the proportion of women with the ovulatory phenotype.⁵⁶ In contrast, Huang et al⁵⁷ found that the SCH distribution was observed similarly in PCOS patients with different phenotypes.⁵⁷

In the case of the hormonal profile, PCOS women with SCH had estradiol levels, which were notably higher compared to euthyroid PCOS patients. However, contrary to our findings, PCOS is generally associated with normal estrogen levels.⁵⁰

The elevated estradiol levels observed in the SCH-PCOS group could potentially be attributed to the previously discussed link between estradiol and the development of autoimmunity. While LH, FSH, and the LH/FSH ratio

typically differ between PCOS and the general population,⁵⁸ only LH levels were significantly higher in SCH-PCOS. This pattern contrasts with obese PCOS patients where hyperinsulinemia often normalizes LH levels, suggesting our findings may be influenced by the metabolic heterogeneity of our cohort. The potential interaction between SCH and this weight-dependent LH secretion warrants further investigation, particularly given SCH's proposed effects on insulin sensitivity.

Disruptions in kisspeptin neurons, influenced by ovarian estrogen, alter GnRH secretion, increasing LH relative to FSH.⁵⁹ However, not all studies consistently report elevated LH levels in PCOS, likely due to phenotypic heterogeneity.⁶⁰⁻⁶³ The variability in PCOS phenotypes is believed to be a key factor contributing to these inconsistent findings in LH, FSH, and LH/FSH ratios across studies. Lean PCOS patients often exhibit higher LH levels and an elevated LH/FSH ratio due to ovarian estrogen disrupting GnRH secretion.^{64,65} Notably, Ganie et al,²³ Enzevaei et al,⁶⁶ and Lu et al⁶⁷ found no significant differences in LH, FSH and LH/FSH ratio. Other studies that examined the LH/FSH ratio similarly reported no notable differences.^{23,57,66,67}

Our results showed no significant differences in BMI ($P = .814$), WHR ($P = .274$) between euthyroid and SCH-PCOS groups as in some studies.^{39,68,69} Trakakis et al⁷⁰ reported no significant differences in BMI and WHR between PCOS patients with SCH and those with normal thyroid function. Similarly, Benetti-Pinto et al⁷¹ provided evidence that aligns with these observations.

Our results showed a significant link between SCH and impaired glucose metabolism, with higher FBG levels and a greater prevalence of impaired glucose tolerance in SCH-PCOS patients. These findings corroborate previous research suggesting that hypothyroid states, even subclinical, can negatively impact insulin sensitivity and glucose metabolism and where SCH and diabetes mellitus were proven to be closely related.⁷² However, Celik et al²⁴ reported that the 120-minute oral glucose tolerance test was comparable among SCH and controls. Pei et al and Trummer et al confirmed these findings.^{25,57} However, Lu et al⁶⁷ observed significant differences at 120 minutes.

While higher total cholesterol levels and a greater prevalence of hypertriglyceridemia were observed in the SCH-PCOS group, compared to euthyroid PCOS patients, the LDL-C, and HDL-C values did not significantly differ. These findings align with studies linking thyroid dysfunction to worsened lipid profiles and increased cardiovascular risk.⁷³

Contrasting with our results, Nazarpour et al⁷⁴ found no significant association between PCOM and SCH (aOR: 1.38, 95% CI: 0.80-2.37) or differences in TSH thresholds between groups, highlighting the need for further population-based studies with detailed thyroid evaluations.

Thyroid hormones regulate lipid metabolism via HMG-CoA reductase and upregulate the LDL receptor gene,⁷⁵ but the exact mechanisms are not yet fully understood. Notably, thyroxine treatment can improve lipid abnormalities within 4 to 6 weeks.^{76,77} The clinical impact of lipid profile alterations remains unclear, particularly regarding long-term cardiovascular risk. PCOS is linked to various endocrine and

metabolic disturbances, all of which elevate the risk of sub-clinical cardiovascular disease.⁷⁸ The combined presence of SCH and PCOS and its potential influence on atherosclerotic cardiovascular disease warrants further investigation, and future studies may offer valuable insights.

A key strength of this study is that it is the first in Tunisia to examine thyroid function, particularly SCH, in relation to PCOS. Beyond assessing their association, it also evaluates SCH's impact on metabolic and hormonal parameters, providing a comprehensive analysis. PCOS diagnosis was based on Rotterdam criteria (two-thirds features required), with ultrasound (transvaginal for sexually active women, abdominal for others) performed only when needed to confirm PCOM in cases lacking hyperandrogenism or oligo/anovulation. Of 161 PCOS patients, 128 (79.5%) met diagnostic criteria without requiring PCOM (hyperandrogenism + ovulatory dysfunction), while 33 (20.5%) were diagnosed via PCOM plus 1 other feature. However, some limitations exist. The small sample size for phenotype D limits generalizability across all PCOS phenotypes. Additionally, as a clinical-based study, selection bias cannot be ruled out. Future research should validate these findings in larger, more diverse populations, including all PCOS phenotypes, to enhance generalizability, and minimize bias.

Conclusion

SCH influenced metabolic and hormonal profiles, potentially worsening insulin resistance and further contributing to the dyslipidemia risk in PCOS. The coexistence of PCOS and thyroid dysfunction may indicate a more severe phenotype with greater reproductive and metabolic risks. These findings highlight the need for closer monitoring of PCOS patients, even in the absence of overt thyroid symptoms. While routine thyroid screening in PCOS may be beneficial, current evidence does not support thyroid hormone supplementation unless further research establishes its metabolic benefits. Future studies with larger, more diverse cohorts are needed to confirm these associations and refine management strategies. Addressing thyroid dysfunction in PCOS could improve reproductive, metabolic, and cardiovascular outcomes, enhancing patient care.

ORCID iD

Taieb Ach  <https://orcid.org/0000-0002-8387-8278>

Ethical Considerations

Ethical approval was obtained from the Medical Ethics Review Committee of Farhat Hached University Hospital in Sousse. The study was conducted in accordance with biomedical ethics principles, ensuring patient respect, confidentiality, and result anonymity.

Consent to Participate

All procedures performed were part of routine medical care. Approval was granted by the Medical Ethics review committee of the Farhat Hached University Hospital in Sousse 250/2024.

Author Contributions

Taieb Ach: Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Resources; Software; Supervision; Validation; Visualization; Writing - original draft; Writing - review & editing. Rim Dhaffar: Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Resources; Software; Supervision; Validation; Visualization; Writing - original draft; Writing - review & editing. Fatma Ben Abdesslem: Conceptualization; Data curation; Methodology; Resources; Supervision. Wiem Saafi: Funding acquisition; Investigation; Supervision; Writing - review & editing. Imen Halloul: Conceptualization; Project administration; Supervision. Hamza ElFekih: Methodology; Supervision; Validation; Writing - review & editing. Ghada Saad: Methodology; Software; Supervision; Writing - review & editing. Yosra Hasni: Project administration; Resources; Supervision; Writing - review & editing.

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